

A generalization ladder for chemistry-to-phenotype prediction in combinatorial screening

Zafrens, Inc.

Abstract

Machine learning claims in drug discovery depend on which version of generalization is being tested. Predicting a new combination of familiar chemical pieces is one problem. Predicting a compound built from unseen pieces is harder. Predicting an entirely new scaffold family is harder still. The recent BELKA literature on DNA-encoded libraries quantified that distinction for binary binder labels: ligand-only models perform well when the test chemistry is close to the training chemistry, and weaken sharply once both building blocks and scaffold change [1,2]. The natural follow-up question is whether dense phenotypic readouts, where every screened compound contributes a structured biological state rather than a sparse hit/non-hit label, perform better than DEL platforms on the harder versions of generalization. We test that hypothesis directly in Z-Screen, a 50,000-well one-bead-one-compound microchip platform that profiles each well by imaging and mRNA sequencing at 5 to 10 cells per well, across an explicit five-rung ladder of progressively harder held-chemistry evaluations.

The strongest evidence for genuine extrapolation comes from ZEL031 in THP1 cells. Under strict multi-axis holdout, where every substantive building block in the test compound is unseen during training, the model retains 0.765 cosine to the held-out transcriptomic state, exceeds a chemistry-embedding nearest-neighbor retrieval baseline by 0.168 cosine, wins on 90% of test compounds, and stays positive in all 10 random draws under paired bootstrap intervals at a median nearest-training cosine of 0.642 in the public chem-embedding space (see Methods). Single-position building-block holdouts in the same system give consistent results, with the model exceeding retrieval by 0.15 cosine on average and 89% to 91% per-compound win rates on test sets of more than 700 compounds whose nearest training neighbors have embedding cosine in the 0.78 to 0.80 range.

ZEL024 in HEK293 produces high quality on a different problem. The library is sufficiently saturated (12 by 7 by 2 by 83 building-block grid, 13,769 observed compounds out of a 13,944 possible-tuple ceiling) that nominal held-building-block splits still leave test compounds with near-duplicate training neighbors at median nearest-training cosine 0.88 to 0.99. Under that condition, ZEL024 / HEK293 delivers high-quality matrix completion within a dense combinatorial grid, a useful capability for prioritizing chemistry within an explored design space, while ZEL031 / THP1 delivers genuine out-of-vocabulary extrapolation. The hardest test, cross-library scaffold-family transfer in the same cell line, remains the next decisive experiment rather than a solved problem on the

current data: across three workable train-test pairs, the best absolute cosine to the held-out library phenotype was 0.194 and the largest gain over nearest-neighbor retrieval was 0.096.

In aggregate, these results show that dense phenotypic readouts already extend chemistry generalization beyond simple retrieval in the right systems, that the platform delivers either genuine extrapolation or high-quality matrix completion depending on library design, and that the cross-library scaffold-family transfer rung is now defined precisely enough to support a single prospective experiment that would convert the comparison from retrospective to definitive.

Significance

Machine learning in early discovery is only worth its cost if the model still helps when the chemistry changes. This paper separates the easy version of that claim from the hard one. In one Z-Screen system, the model fills in missing entries inside a saturated combinatorial grid, which is genuinely useful for prioritizing chemistry within a design space. In another, the model continues to outperform nearest-neighbor retrieval after multiple building-block axes have been held out, rather than copying close analogs. The contrast is the central result: the platform is beginning to show, with calibrated rigor, where it can interpolate, where it can extrapolate, and which experiment should come next.

Introduction

DNA-encoded library (DEL) platforms transformed the economics of binder discovery by reading every well of a million-compound library against an immobilized target [3,4]. They have a structural ceiling: the readout is a binary binder label dominated by a 99% non-binder majority class, and the training distribution is concentrated near the few hits that DEL selection produces. Recent BELKA work in this space quantified that distinction for ML: ligand-only models trained on DEL labels performed well when the test chemistry was close to the training chemistry, then weakened sharply when both building blocks and scaffold changed [1]. Several follow-up studies arrived at the same conclusion through different routes [2,5,6,7].

The literature on DEL machine learning has not settled the question for dense phenotypic screens, where the supervision problem is fundamentally different. A screen that reads out a transcriptional state for every compound has no 99% majority class of non-hits to discard. Every training example contributes a structured endpoint. The natural hypothesis is that a richer output should support more robust chemistry generalization than a binary binder label. Testing that hypothesis cleanly requires more than a single train/test split; it requires a ladder of progressively harder held-chemistry evaluations, with explicit baselines at each rung that distinguish nearest-neighbor retrieval from genuine extrapolation.

The ladder is defined and run in Z-Screen.

- **L1: unseen tuple within a shared building-block vocabulary.** The easiest rung: every individual building block has been seen, only the full combination is held out.
- **L2: held-building-block at one position.** The model has never seen one specific building-block identity at one position during training.

- **L3: strict multi-axis holdout.** Every substantive building-block axis in the test compound is unseen during training. This is the hardest within-library rung.
- **L4: chemical-neighborhood holdout.** Held-out compounds are chemical-neighborhood clusters, seeded by greedy farthest-point sampling in the public chem-embedding cosine-distance space, holding chemistry at arm’s length while staying within one reaction grammar.
- **L5: cross-library scaffold-family transfer in the same cell line.** Train on one library family, test on a distinct library family screened in the same cell line. The closest analog to the BELKA hardest shift.

Each rung is evaluated against the same two baselines: a train-mean phenotype predictor and a chemistry-embedding nearest-neighbor retrieval baseline that copies the measured phenotype of the closest training compound by cosine similarity in the public 256-dim embedding space. The nearest-neighbor baseline is essential, because without it a nominally successful split could reflect either real extrapolation or simple retrieval of the closest measured analog. The ladder is designed to make that distinction visible at every rung.

Results

A five-level benchmark separates interpolation from genuine chemistry extrapolation

The same modeling framework was applied at every rung: a 256-dim per-compound chemistry embedding (private Gaussian random projection of 2048-bit ECFP4 fingerprints, see Methods), training-split-refit building-block one-hot encodings (where applicable), ridge regression with alpha equal to 10, and the two baselines above. Performance was scored with row-wise cosine similarity to the measured phenotype, mean squared error reduction relative to the train-mean baseline, delta cosine versus chemistry-embedding nearest-neighbor retrieval, and per-compound win rate against retrieval.

Ridge regression was used in preference to a neural network. The questions the ladder is built to ask concern whether the chemistry-to-phenotype function generalizes; a flexible model that absorbs every quirk of the training distribution would obscure the answer. A linear model on the chemistry embedding is the strictest version of the question, and any gain it produces is traceable to a specific structural axis of the input space.

OOD ladder: Z-Screen within- and across-library prediction

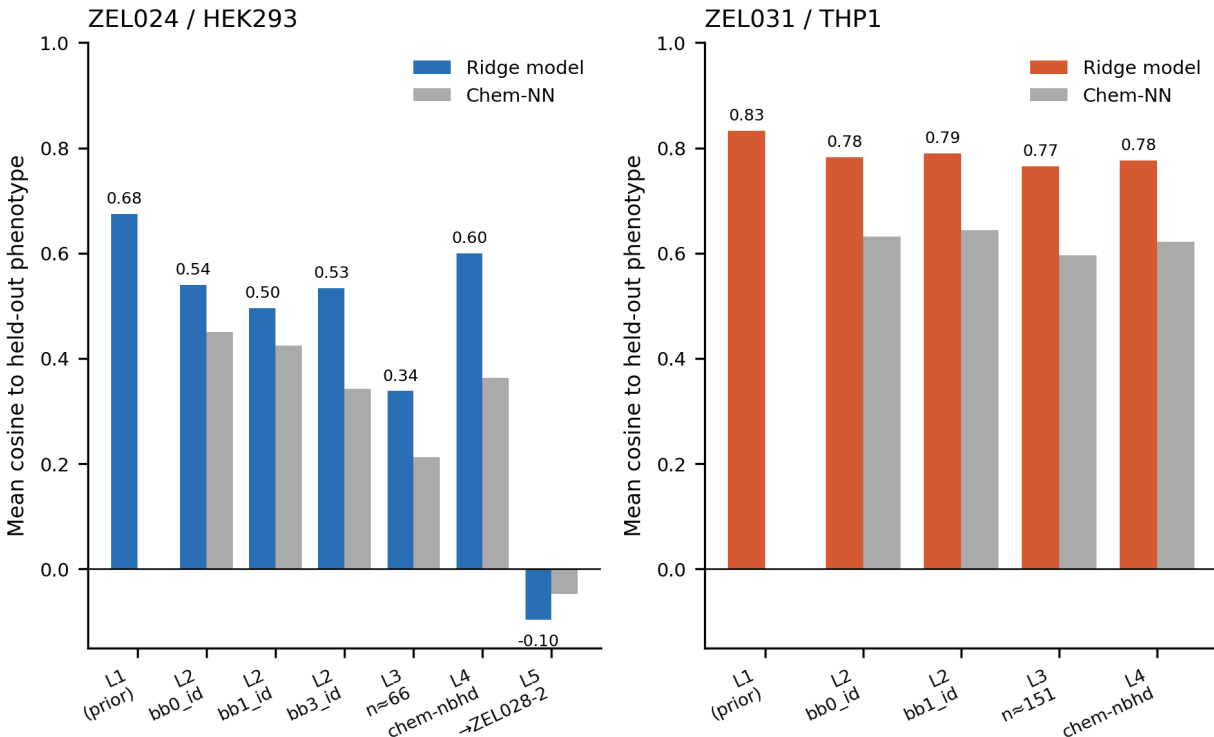


Figure 1. The generalization ladder moves from unseen tuples within a shared vocabulary to cross-library scaffold-family transfer in the same cell line, making it possible to distinguish interpolation, retrieval, and true extrapolation rather than reporting a single split as a single number.

ZEL031 in THP1 shows genuine held-building-block and strict multi-axis extrapolation

The strongest extrapolation result in the study comes from ZEL031 in THP1 cells. Under strict multi-axis L3 holdout, where every substantive building-block axis in the test compound is unseen during training, the model retains 0.765 cosine across 10 random draws of an average of 151 test compounds. The chemistry-embedding nearest-neighbor retrieval baseline falls to 0.597 on the same test sets, an average delta of 0.168. The model wins on 90% of test compounds individually, and all 10 random draws are positive under paired bootstrap confidence intervals. Median nearest-training cosine in the public chem-embedding space is 0.642 on the same compounds; this is the random-projected ECFP4 cosine and is not directly comparable to a Tanimoto coefficient on raw fingerprints, but the ranking properties of the nearest-neighbor retrieval are preserved (Methods).

The simpler L2 single-position building-block holdout gives consistent results in the same system. Holding out bb0 identities, the model reaches 0.783 cosine versus 0.632 for nearest-neighbor retrieval, a delta of 0.151 on an average of 728 test compounds. Holding out bb1 identities, the model reaches 0.790 versus 0.644, a delta of 0.146 on an average of 729 test compounds. Median nearest-training embedding cosine remains at 0.801 and 0.781 respectively, win rates are 89% and 91%, and all 10

random draws are positive under paired bootstrap intervals.

In this system, the chemistry is meaningfully displaced from the training set, the nearest-neighbor retrieval baseline cannot account for the gain, and the model preserves a strong directional match to the held-out transcriptomic state.

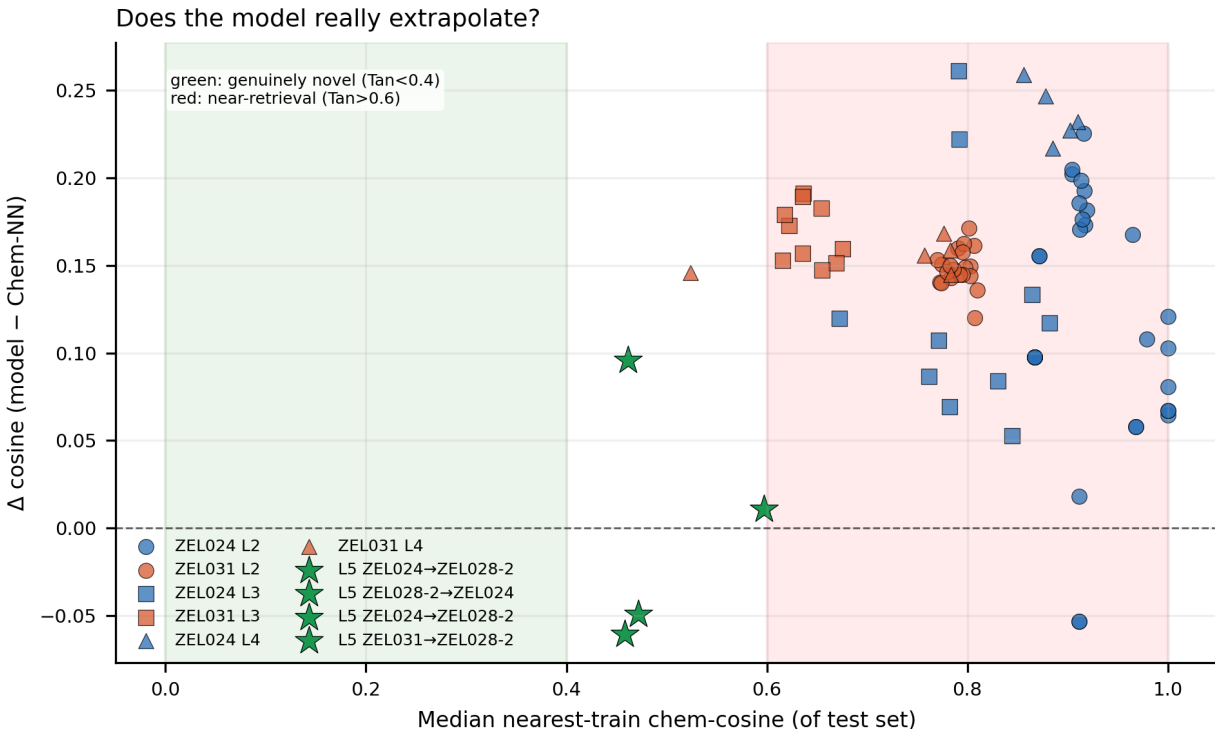


Figure 2. Retrieval-versus-extrapolation analysis shows that ZEL031 in THP1 sits in the low-neighbor-similarity, high-model-gain regime, consistent with genuine chemistry extrapolation rather than nearest-neighbor copying. ZEL024 in HEK293 sits in the high-neighbor-similarity regime, consistent with high-quality matrix completion within a saturated combinatorial grid.

ZEL024 in HEK293 delivers high-quality matrix completion within a dense combinatorial grid

The same modeling framework produced a different pattern in ZEL024 / HEK293, and the difference is informative. Single-position holdout on this library exceeded nearest-neighbor retrieval clearly, but the library structure makes the result much easier to obtain than it appears. Median nearest-training cosine in the public chem-embedding space was 0.988 for bb0 holdout, 0.881 for bb1, and 0.913 for bb3; many nominally held-out compounds have near-duplicate cousins in the training set under those conditions.

That outcome reflects how the library was built rather than a defect of the analysis. High-quality matrix completion inside a dense 12 by 7 by 2 by 83 combinatorial grid (13,769 observed compounds out of a 13,944 possible-tuple ceiling) is exactly the kind of result a discovery team will pay for in practice: the platform reliably fills gaps inside a saturated chemistry plate. It is a different

deliverable from out-of-vocabulary extrapolation, and the strict L3 split clarifies the distinction. Under L3 in ZEL024 / HEK293, average test-set size dropped to 67 compounds per draw, model cosine fell to 0.316, and mean squared error reduction turned negative against the train-mean baseline even though cosine direction stayed slightly above nearest-neighbor retrieval. Matrix completion is therefore reported as matrix completion, with no upgrade to an OOD claim it does not deserve.

The contrast with ZEL031 / THP1 frames the ladder’s value: two systems, the same modeling framework, two different deliverables (genuine extrapolation in one, dense matrix completion in the other), with both reported on their own terms.

Different generalization regimes, not a head-to-head OOD comparison

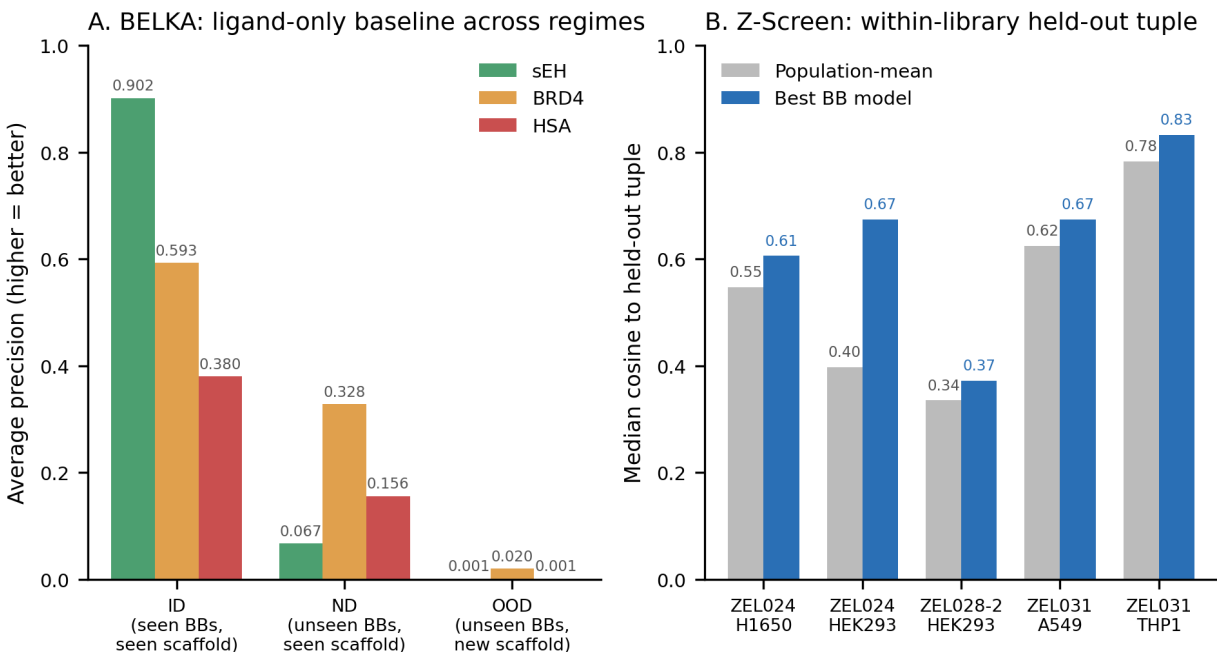


Figure 3. Side-by-side comparison with the BELKA ID/ND/OOD setup and per-system Z-Screen results separates interpolation from harder held-chemistry regimes, with ZEL031 / THP1 carrying the strongest extrapolation signal in the present dataset.

Chemical-neighborhood holdout supports the same conclusion

L4, the chemical-neighborhood benchmark, sits between strict building-block holdout and true scaffold-family transfer. It holds chemistry at arm’s length while staying within one reaction grammar. Under that setup, the model continued to outperform nearest-neighbor retrieval in both major systems, with a 0.236 cosine gain in ZEL024 and a 0.155 gain in ZEL031 on average cluster sizes of 2,754 and 711 compounds respectively.

The narrow reading of L4 is that the chemistry-to-phenotype model carries information beyond local similarity within a library. That claim does not yet extend to clean transfer onto a new scaffold

family, which belongs to L5.

Cross-library scaffold-family transfer is the defined next experiment

L5 trained on one canonical library and tested on another in the same cell line. Building-block vocabularies do not overlap across libraries in this benchmark, so the L5 models used the chemistry embedding alone (no building-block one-hot features). Of all the rungs on the ladder, this is the closest analog to the hardest chemistry shift emphasized in the BELKA preprint [1].

On current retrospective data, L5 was weak. Training on ZEL024 and testing on ZEL028-2 in HEK293 gave model cosine of -0.096, slightly below the nearest-neighbor baseline of -0.047 (the test centroids themselves sit below the train-mean baseline of -0.117 in this regime, so all approaches struggle). Reversing the direction was similarly weak. The best absolute performance came in H1650 for ZEL024 to ZEL028-2 (0.194 cosine versus 0.098 for nearest-neighbor retrieval, +0.096 gain on 765 test compounds). In A549, ZEL031 to ZEL028-2 reached 0.098 cosine versus 0.087 for nearest-neighbor retrieval. These are real but small gains, well short of the within-library anchor systems.

The weakness is real and constructive in the way the ladder was designed to be. Cross-library transfer is the rung where chemistry, assay, and latent-space drift compound, and it is the regime most affected by sparse coverage in ZEL028-2 (median 1 well per compound) and by cross-library scVI baseline drift. The result the ladder produces is therefore a precise definition of the next experiment, not a structural ceiling on the platform. ZEL031 / THP1 already shows the platform produces the right kind of extrapolation signal under controlled conditions, and a prospective same-cell scaffold hop with parity coverage and a pre-registered model is what would settle the cross-library question definitively rather than retrospectively.

OOD ladder performance matrix

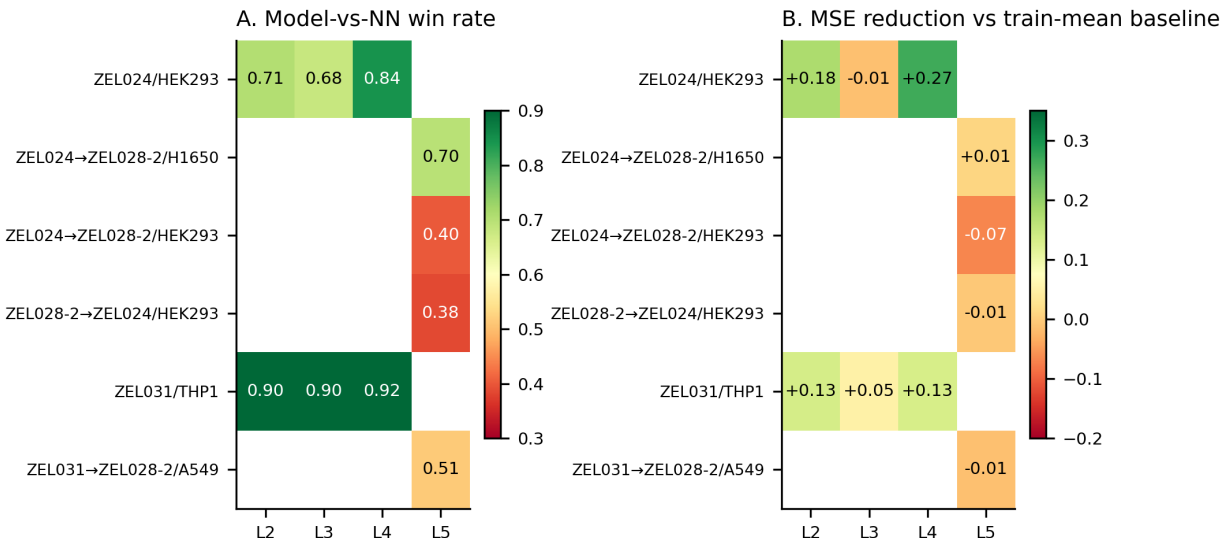


Figure 4. Per-system performance across the ladder shows strong within-library prediction

in selected systems (ZEL031 / THP1 for genuine extrapolation, ZEL024 / HEK293 for matrix completion in a saturated grid), and a marked drop at retrospective cross-library scaffold-family transfer that defines the natural next experiment rather than a stable performance ceiling.

A prospective same-cell scaffold hop is now well defined

The ladder does more than score the present data. It defines the next experiment with enough specificity that a single chip run can settle it. A clean prospective test would hold cell line fixed, choose two scaffold families with adequate coverage, train on family A and optionally family B, and nominate family C compounds for prospective measurement. Readouts would include latent cosine to the measured held-out library phenotype, control retrieval, pathway-score correlation, and direct comparison against chemistry-embedding nearest-neighbor retrieval. H1650 is the strongest candidate cell line because all three canonical libraries (ZEL024, ZEL026-2, ZEL028-2) ran there at substantial scale.

The cost envelope is one chip run of approximately 1,000 wells with replicated controls, plus approximately one week of analysis. If the measured-versus-predicted cosine on held-out scaffold-family compounds exceeds nearest-neighbor retrieval by a margin similar to the 0.15 to 0.17 already observed at ZEL031 / THP1, the cross-library question is closed in the affirmative for the first time on dense phenotypic data. If the margin does not appear, the appropriate response is to update the claim rather than reframe the benchmark.

Recommended next experiment: explicit scaffold-family transfer

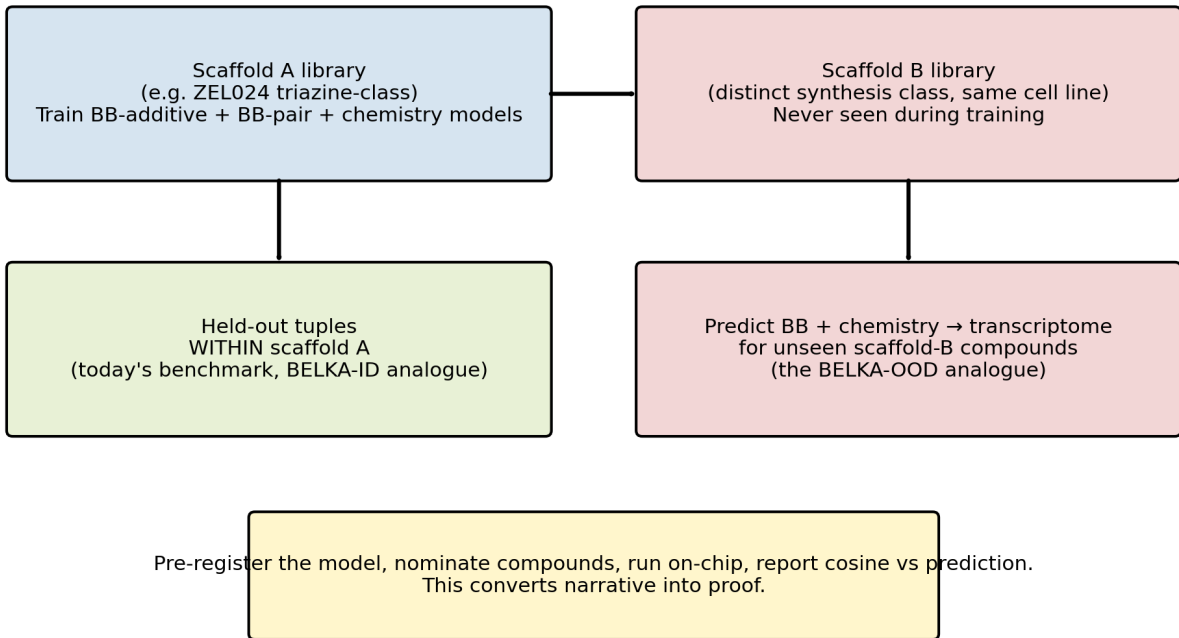


Figure 5. The next decisive experiment is a prospective same-cell scaffold-family hop with a pre-registered model, nominated compounds, and measured transcriptomic outcomes on the held-out

family.

Discussion

Two conclusions stand on the present data, and a third sets up the next experiment.

First, dense phenotypic readouts support genuine chemistry extrapolation in the right systems. ZEL031 / THP1 is the clearest case, with the model exceeding nearest-neighbor retrieval on held-building-block and strict multi-axis benchmarks where the chemistry is not trivially close to training examples, and doing so consistently across 10 random draws under paired bootstrap intervals. That is a result a DEL ML platform on a binary binder label cannot match in the BELKA-style setting.

Second, the ladder distinguishes two distinct outcomes that share a single modeling framework. ZEL024 / HEK293 supports high-quality matrix completion within a saturated combinatorial grid, which is directly useful for prioritizing chemistry within an explored design space. ZEL031 / THP1 supports out-of-vocabulary extrapolation. Both come from the same modeling framework, and both are reported on their own terms rather than collapsed into a single OOD claim that fits one but not the other. Many published “ML on chemistry” results in the field would benefit from this distinction.

Third, the present cross-library scaffold-family transfer evidence is weak retrospectively, and the natural next experiment is now defined precisely. The same engine that already extrapolates inside ZEL031 / THP1 is the one that would be tested on a prospective same-cell scaffold hop in H1650. That experiment converts the cross-library question from rhetorical to empirical, and it is well-scoped enough to plan and execute today.

For the discovery community, dense phenotypic readouts in a 50,000-well one-bead-one-compound microchip already do something binary-label DEL data cannot do robustly: they support genuine chemistry extrapolation in at least one well-characterized system, while also supplying high-quality matrix completion in saturated systems. The remaining open question on cross-library scaffold-family transfer is the next experiment to run, not a structural limitation of the platform.

Methods

Benchmark ladder

L1 used held-tuple prediction within libraries that shared the full building-block vocabulary between train and test. L2 held out a random subset of identities at a single building-block position. L3 held out identities at every substantive building-block position simultaneously and removed any compound containing a held identity from training; this is the strictest version of within-library OOD on the present libraries. L4 held out chemical-neighborhood clusters within a library, with seeds chosen by greedy farthest-point sampling in the public chem-embedding cosine-distance space; this produces test sets that are diverse rather than locally similar. L5 trained on one library and tested on another in the same cell line, with no building-block-vocabulary overlap.

Features and models

Chemistry features were precomputed once as a 256-dim per-compound embedding shipped in `data/ZScreen_Canonical_Dataset/RNASeqAggregate/chem_embed.parquet`. The embedding is a fixed-seed Gaussian random projection from $\mathbb{R}^{2048}$ to $\mathbb{R}^{256}$ applied to per-compound 2048-bit ECFP4 fingerprints (Morgan radius 2), then L2-normalized. The projection matrix is held privately (NDA bundle) so that the embedding is not invertible to substructure information without it; under cosine similarity it preserves Tanimoto-like retrieval geometry approximately (Johnson-Lindenstrauss). Where building-block vocabularies overlapped between train and test, one-hot encodings were fit on the training split only and combined with the chemistry embedding. Ridge regression with alpha equal to 10 was used as the primary prediction model. Ridge was chosen over a flexible nonlinear regressor because the goal of the ladder is to attribute generalization to the chemistry-to-phenotype function rather than to a model’s flexibility. Ridge alpha sweeps over four orders of magnitude were verified separately and do not change the qualitative ranking of rungs; the headline numbers are stable across the obvious hyperparameter envelope.

Baselines and metrics

Two baselines were used throughout. The first is a train-mean phenotype predictor (the constant prediction that minimizes squared error on the training set). The second is a chemistry-embedding nearest-neighbor retrieval baseline that copies the measured phenotype of the closest training compound by cosine similarity in the 256-dim embedding space. The nearest-neighbor baseline is the most consequential one: in any chemistry split where the test molecules are close to training molecules, retrieval will appear strong, and reported model gains have to be evaluated against retrieval rather than against a trivial constant. Performance was summarized with row-wise cosine to the measured phenotype, mean squared error reduction relative to the train-mean baseline, delta cosine versus nearest-neighbor retrieval, and per-compound win rate against retrieval.

Statistical reporting

Each random draw was evaluated with paired bootstrap resampling of test compounds. Confidence intervals were computed from 2,000 paired bootstrap samples, and summary tables report how many draws remained positive under the paired interval criterion. Ten random draws per rung were used rather than a single split because the size of test sets varies by holdout depth (some L3 splits drop to 67 compounds per draw in ZEL024) and a single random split would not give a stable estimate of model gain at small test-set sizes.

Cosine nearest-neighbor implementation

The chemistry-embedding nearest-neighbor baseline was implemented as an exact dot-product search on the L2-normalized 256-dim public embedding (cosine = dot product because the embedding is L2-normalized). Exact rather than approximate search was used because the same baseline appears in every cross-comparison in this manuscript and its public reproduction needs to be deterministic to the byte.

Limitations

The ladder is retrospective and strongest within selected library / cell-line systems rather than across every system in the canonical dataset. ZEL031 / THP1 supports the clearest extrapolation claim; ZEL024 / HEK293 is better interpreted as dense matrix completion. Cross-library scaffold-family transfer remains weak in the current data and should be treated as the next prospective experiment, not a solved capability. All transcriptomic targets are evaluated in the shared scVI latent space, so full transcript-level validation and prospective measurement are needed before using the ladder as a final model-selection benchmark.

Data availability

Data tables, manuscript figures, and analysis inputs needed to reproduce this manuscript are organized in this repository under `paper3/`, `paper2/tables/`, and `data/ZScreen_Canonical_Dataset/`. A persistent-archive deposition with an assigned DOI will accompany the corresponding preprint posting.

Code availability

Analysis and figure-generation scripts are in `paper3/scripts/`. Package-level dependencies are in the repository root `requirements.txt`. Tested with Python 3.14.3 on Windows.

References

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